

SENTINELGUARD

SOC Monitoring, Identity Security & Risk Management Platform

SECURITY OPERATIONS CENTER — FINAL REPORT

Classification: CONFIDENTIAL

Report Date: June 18, 2026

Prepared by: SOC Analyst — Nimesh Nilakshan

Target System: DESKTOP-16UFKJN (192.168.56.1)

Incident ID: INC-2026-001

Contents

1. Executive Summary	3
2. Lab Environment & Tools Used	4
2.1 Architecture -:.....	4
2.2 Tools Used -:.....	5
3. Identity & Access Management.....	5
3.1 Simulated Employee Dataset -:.....	5
4. Endpoint Monitoring Configuration	7
4.1 Log Sources Configured	7
4.2 Key Event IDs Monitored.....	8
5. Password Security Audit	8
5.1 Password Audit Results:.....	9
5.2 Key Finding:.....	9
6. Threat Detection — Splunk Alerts.....	10
7. Attack Simulation	12
7.1 Attack Details:.....	12
7.2 Attack Events Generated in Splunk	13
8. Incident Investigation.....	13
8.1 Incident Summary:	13
8.2 Indicators of Compromise (IOCs):.....	14
9. Risk Assessment.....	15
9.1 Per-Account Risk Rating:.....	15
9.2 Risk Scoring Criteria:	16
10. Recommendations	17
10.1 Immediate Actions Required:	17
10.2 Short-Term Improvements (Within 30 Days):	17
10.3 Long-Term Security Improvements:.....	17
11. Conclusion.....	18

1.Executive Summary

This report documents the findings of a Security Operations Centre (SOC) assessment conducted on the SentinelGuard simulated enterprise environment. The assessment covered endpoint monitoring, identity and access management, password security auditing, threat detection, attack simulation, incident investigation, and risk assessment.

A brute-force attack was simulated from a Kali Linux machine targeting all five simulated employee accounts via SMB (port 445). The attack generated 120 failed login events across all targeted accounts. No successful unauthorized login was confirmed. Two employee accounts were identified as critically vulnerable due to weak, easily crackable passwords.

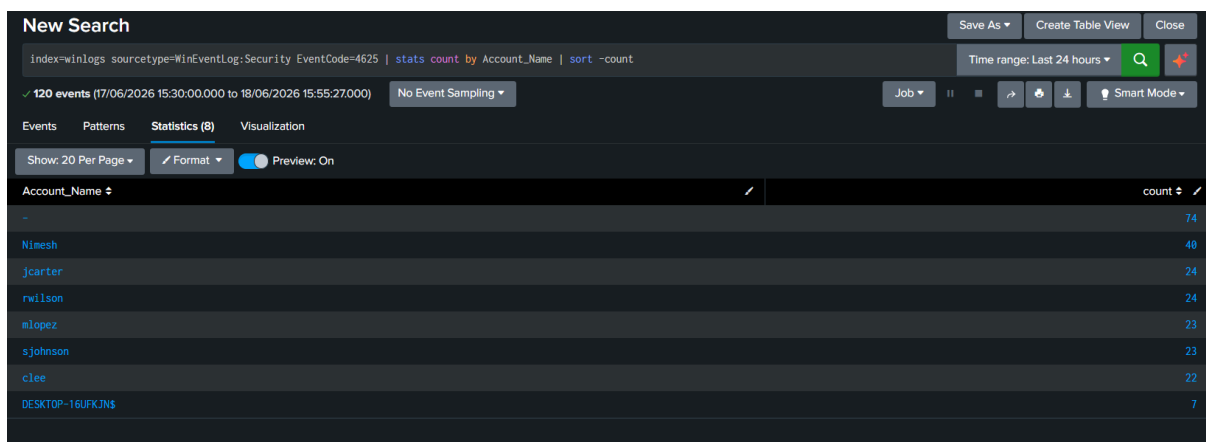


Figure 1: Failed Login Count

Metric		Value
Assessment Date		June 18, 2026
Target System		DESKTOP-16UFGJN (192.168.56.1)
Total Accounts Audited		5 employee accounts
Total Failed Login Events		120
Successful Unauthorized Logins		0 — No breach confirmed
Passwords Cracked		2 of 5 (jcarter, rwilson)
Overall Risk Rating		HIGH

2. Lab Environment & Tools Used

2.1 Architecture -:

Host Machine: Windows 11 (real host) — Target system running Splunk Enterprise, Sysmon, and Windows Security Auditing.

Attack Machine: Kali Linux (VirtualBox VM, Host-Only Adapter, 192.168.56.x subnet).

Communication: Isolated Host-Only virtual network — no exposure to home/external network.

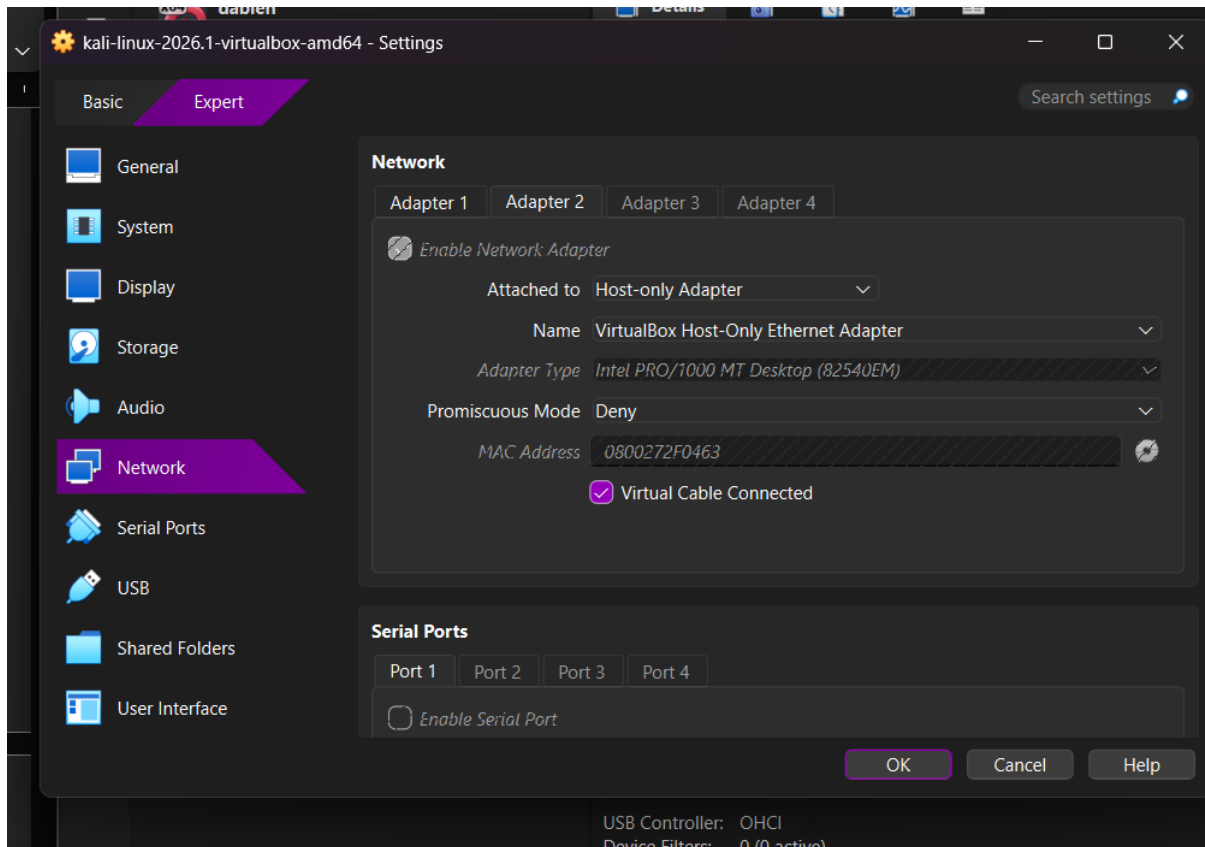


Figure 2: VirtualBox Network

2.2 Tools Used -:

Tool	Purpose	Location
Splunk Enterprise	SIEM — Log ingestion, search, alerting	Windows Host
Sysmon	Endpoint telemetry (process, network, file)	Windows Host
netexec	SMB brute-force attack simulation	Kali Linux
John the Ripper	Password hash cracking audit	Kali Linux
OpenSSL	SHA-512 password hash generation	Kali Linux
Windows Event Logs	Security, System, Application logging	Windows Host

3. Identity & Access Management

3.1 Simulated Employee Dataset -:

A dataset of 12 simulated employees was created across 5 departments to represent a realistic small-company environment. Five of these accounts were created as real local Windows accounts for attack simulation and log generation purposes

	A	B	C	D	E	F	G	H
1	Employee_ID	Full Name	Department	Username	Email	Job Title	Date Hired	
2	EMP001	John Carter	IT	jcarter	jcarter@sentinelguard-corp.local	Systems Administrator	3/14/2022	
3	EMP002	Maria Lopez	IT	mlopez	mlopez@sentinelguard-corp.local	Network Engineer	7/1/2023	
4	EMP003	David Kim	IT	dkim	dkim@sentinelguard-corp.local	Help Desk Technician	1/22/2024	
5	EMP004	Sarah Johnson	Finance	sjohnson	sjohnson@sentinelguard-corp.local	Accountant	11/9/2021	
6	EMP005	Michael Chen	Finance	mchen	mchen@sentinelguard-corp.local	Financial Analyst	2/17/2023	
7	EMP006	Emily Davis	Finance	edavis	edavis@sentinelguard-corp.local	Accounts Payable Clerk	9/30/2022	
8	EMP007	Robert Wilson	HR	rwilson	rwilson@sentinelguard-corp.local	HR Coordinator	5/12/2023	
9	EMP008	Linda Martinez	HR	lmartinez	lmartinez@sentinelguard-corp.local	HR Manager	6/25/2021	
10	EMP009	James Anderson	Sales	janderson	janderson@sentinelguard-corp.local	Sales Executive	2/3/2024	
11	EMP010	Patricia Thomas	Sales	pthomas	pthomas@sentinelguard-corp.local	Sales Representative	10/19/2023	
12	EMP011	Christopher Lee	Operations	clee	clee@sentinelguard-corp.local	Operations Analyst	12/8/2022	
13	EMP012	Jennifer White	Operations	jwhite	jwhite@sentinelguard-corp.local	Operations Coordinator	8/14/2023	
14								
15								

Figure 3: employees.csv file

Account	Full Name	Department	Job Title	Risk Level
jcarter	John Carter	IT	Systems Administrator	HIGH
rwilson	Robert Wilson	HR	HR Coordinator	CRITICAL
mlopez	Maria Lopez	IT	Network Engineer	MEDIUM
sjohnson	Sarah Johnson	Finance	Accountant	MEDIUM
clee	Christopher Lee	Operations	Operations Analyst	MEDIUM

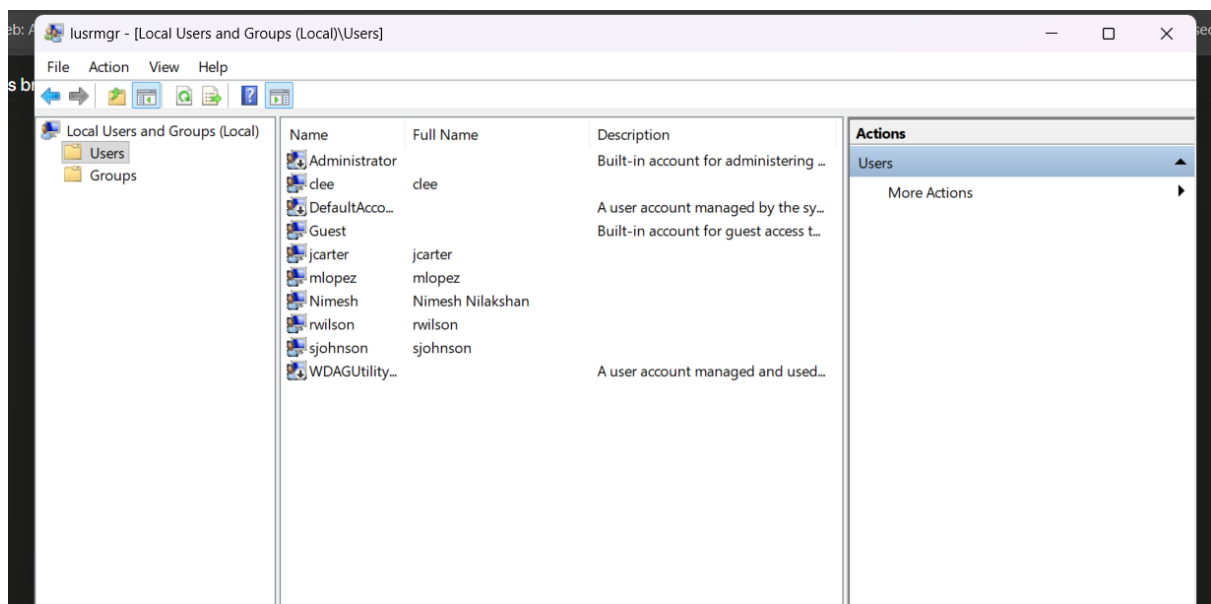


Figure 4: Windows Local Users

4. Endpoint Monitoring Configuration

Windows 11 was configured with Sysmon and Windows Security Auditing to generate detailed telemetry. All log sources were forwarded into Splunk Enterprise for centralized analysis.

4.1 Log Sources Configured

Source	Splunk Sourcetype	Events Collected
Windows Security Log	WinEventLog:Security	9000+
Windows System Log	WinEventLog:System	600+
Windows Application Log	WinEventLog:Application	400+
Sysmon Operational	xmlwineventlog	37000+

The screenshot shows the Splunk Search interface with the following details:

- Search Bar:** `index=winlogs | stats count by sourcetype`
- Time Range:** Last 24 hours
- Results:** 48,668 events (17/06/2026 22:30:00.000 to 18/06/2026 23:15:26.000)
- Tab:** Statistics (4)
- Table:**

sourcetype	count
WinEventLog:Application	433
WinEventLog:Security	9808
WinEventLog:System	602
xmlwineventlog	37833

Figure 5: showing all 4 sourcetypes with their event counts

4.2 Key Event IDs Monitored

- EventCode 4624 — Successful logon
- EventCode 4625 — Failed logon attempt
- EventCode 4672 — Special/elevated privileges assigned
- Sysmon Event 1 — Process creation
- Sysmon Event 3 — Network connection

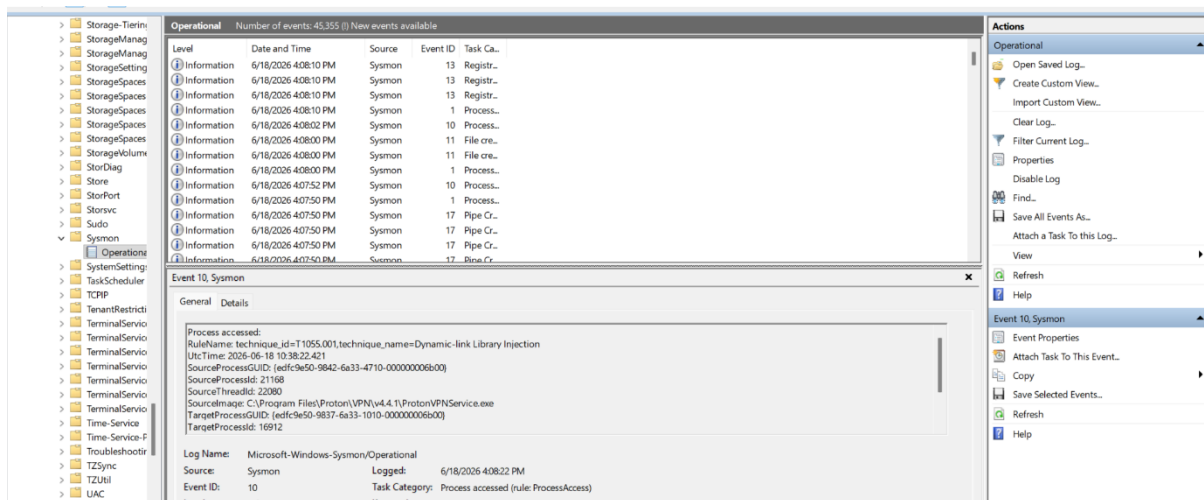


Figure 6: showing live Sysmon events

5. Password Security Audit

A password security audit was performed using John the Ripper against SHA-512 hashed versions of the 5 employee account passwords. The rockyou.txt wordlist (14+ million entries) was used as the attack dictionary

```
(kali@kali)~/sentinelguard
$ john --show ~/sentinelguard/password_audit/hashes.txt
jcarter:Password123
rwilson:123456
2 password hashes cracked, 3 left
```

Figure 7: output showing jcarter

5.1 Password Audit Results:

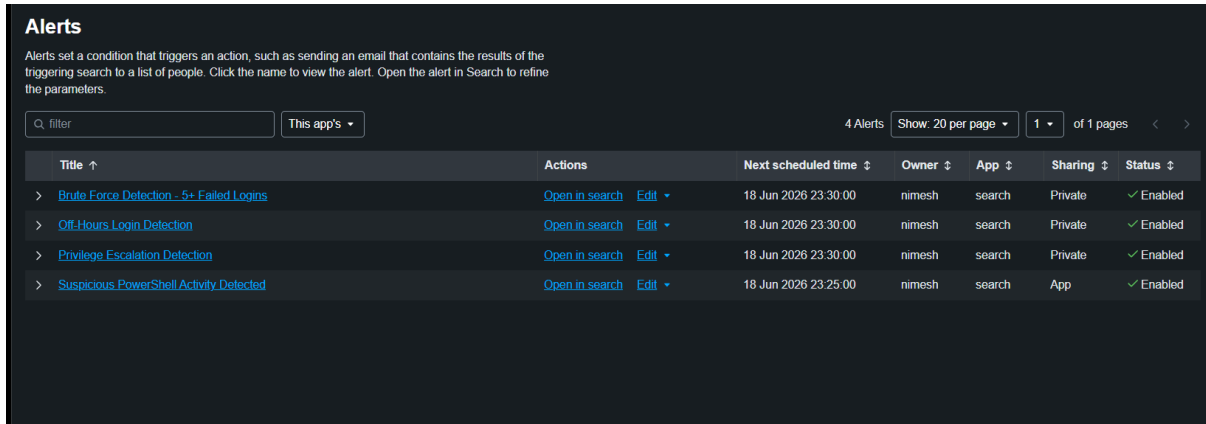
Account	Password	Length	Upper	Lower	Num	Special	Cracked ?
Jcarter	Password123	11	Yes	Yes	Yes	No	YES
Rwilson	123456	6	No	No	Yes	No	YES
Mlopez	Summer2024!	11	Yes	Yes	Yes	Yes	No
Sjohnson	Tr0ub4dor&3xQp9!	16	Yes	Yes	Yes	Yes	No
Clee	K7#mP9\$vL2qX	12	Yes	Yes	Yes	Yes	No

5.2 Key Finding:

2 of 5 accounts (40%) had passwords crackable within seconds using a standard dictionary attack. rwilson used '123456' one of the most common passwords globally. jcarter used 'Password123' a predictable dictionary word with number suffix. Both would be cracked in a real-world attack within minutes.

6. Threat Detection — Splunk Alerts

Three automated detection alerts were configured in Splunk to monitor for active threats in real time. All alerts run on a 15-minute Cron schedule.



Title ↑	Actions	Next scheduled time ↕	Owner ↕	App ↕	Sharing ↕	Status ↕
> Brute Force Detection - 5+ Failed Logins	Open in search Edit ▾	18 Jun 2026 23:30:00	nimesh	search	Private	✓ Enabled
> Off-Hours Login Detection	Open in search Edit ▾	18 Jun 2026 23:30:00	nimesh	search	Private	✓ Enabled
> Privilege Escalation Detection	Open in search Edit ▾	18 Jun 2026 23:30:00	nimesh	search	Private	✓ Enabled
> Suspicious PowerShell Activity Detected	Open in search Edit ▾	18 Jun 2026 23:25:00	nimesh	search	App	✓ Enabled

Figure 8: Splunk Alerts

Alert Name	Detection Logic	Severity	Schedule
Brute Force Detection	5+ failed logins (EventCode=4625) within 5-min window	HIGH	Every 15 min
Privilege Escalation Detection	EventCode=4672 for non-system accounts	HIGH	Every 15 min
Off-Hours Login Detection	EventCode=4624 outside 09:00–18:00 for real users	MEDIUM	Every 15 min

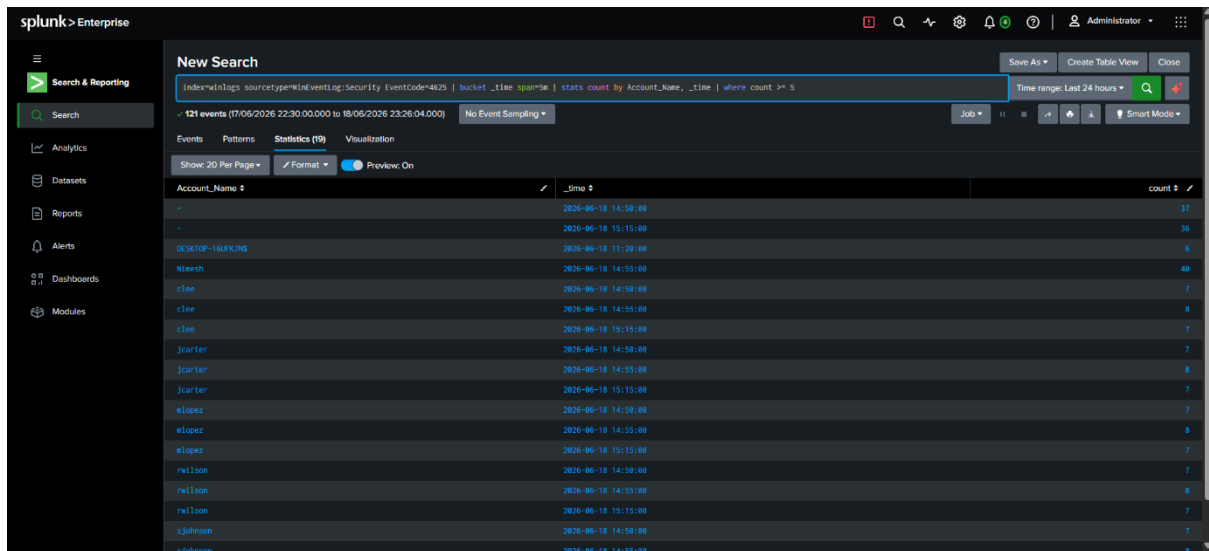


Figure 9: Splunk Search showing brute force threshold

7. Attack Simulation

7.1 Attack Details:

Field	Detail
Attack Type	SMB Brute-Force Authentication Attack
Tool	netexec v1.x (Kali Linux)
Source	Kali Linux VM (VirtualBox Host-Only Network)
Target	192.168.56.1 — DESKTOP-16UFKJN (Windows 11)
Port	TCP 445 (SMB — Microsoft Directory Services)
Accounts Targeted	jcarter, mlopez, sjohnson, rwilson, clee
Wordlist	7 common passwords per account = 35 total attempts
Date/Time	2026-06-18 starting 11:24
Outcome	STATUS_LOGON_FAILURE on all attempts — no breach

```
(kali@kali) ~/sentinelguard
$ netexec rdp 192.168.56.1 -u usernames.txt -p passwords.txt

(kali@kali) ~/sentinelguard
$ netexec smb 192.168.56.1 -u usernames.txt -p passwords.txt --smb-timeout 10
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] Windows 11 / Server 2025 Build 26100 x64 (name:DESKTOP-16UFKJN) (domain:DESKTOP-16UFKJN) (signing:False) (SMBv1:None)
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\jcarter:123456 STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\mlopez:123456 STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\sjohnson:123456 STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\rwilson:123456 STATUS_PASSWORD_MUST_CHANGE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\clee:123456 STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\jcarter:password STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\mlopez:password STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\sjohnson:password STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\rwilson:password STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\clee:password STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\jcarter:admin123 STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\mlopez:admin123 STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\sjohnson:admin123 STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\rwilson:admin123 STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\clee:admin123 STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\jcarter:Password123 STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\mlopez:Password123 STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\sjohnson:Password123 STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\rwilson:Password123 STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\clee:Password123 STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\jcarter:qwerty STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\mlopez:qwerty STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\sjohnson:qwerty STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\rwilson:qwerty STATUS_LOGON_FAILURE
SMB 192.168.56.1 445 DESKTOP-16UFKJN [!] DESKTOP-16UFKJN\clee:qwerty STATUS_LOGON_FAILURE
```

Figure 10: netexec output

7.2 Attack Events Generated in Splunk

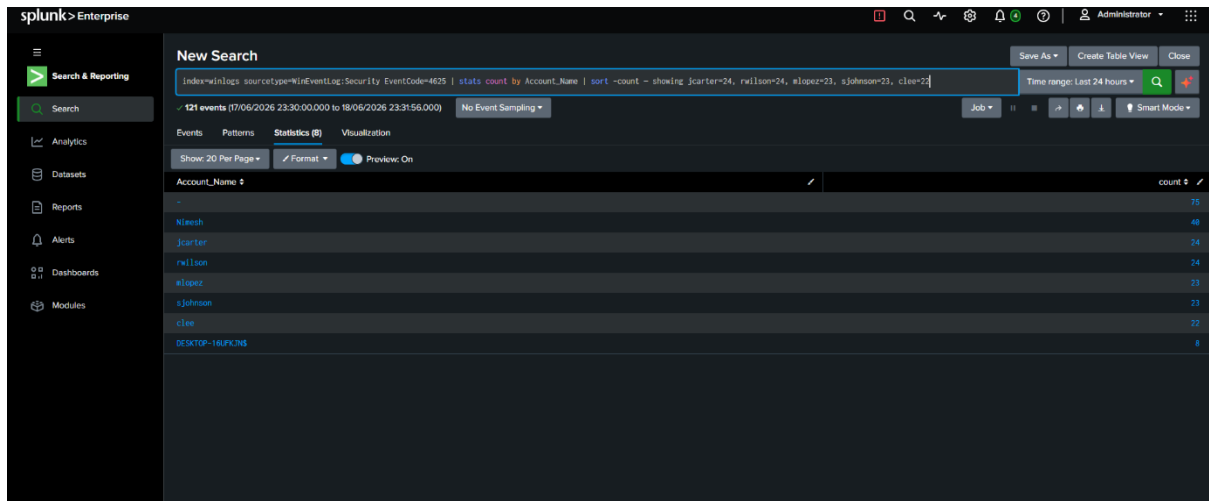


Figure 11: showing failed login counts

8. Incident Investigation

8.1 Incident Summary:

Field	Detail
Incident ID	INC-2026-001
Detection Method	Splunk Brute Force Detection Alert
Attack Start Time	2026-06-18 11:24
Attack End Time	2026-06-18 11:25 (approx)
Affected Accounts	jcarter, mlopez, sjohnson, rwilson, cleo
Total Failed Logins	120 (across all accounts)
Successful Logins Detected	None — no breach confirmed
Privilege Escalation	None detected on employee accounts
IOC	Rapid sequential SMB login failures across all accounts

New Search

index=winlogs sourcetype=WinEventLog:Security (EventCode=4625 OR EventCode=4624 OR EventCode=4672) | table _time, EventCode, Account_Name | sort _time

Time range: Last 24 hours

798 events (7/06/2026 23:30:00.000 to 18/06/2026 23:42:03.000) No Event Sampling

Events Patterns Statistics (798) Visualization

Show: 20 Per Page Format Preview: On

_time	EventCode	Account_Name
2026-06-18 03:21:50.585	4672	SYSTEM
2026-06-18 03:21:50.585	4624	DESKTOP-16UPKJNS
2026-06-18 03:21:50.585	4624	SYSTEM
2026-06-18 03:21:50.585	4624	DESKTOP-16UPKJNS
2026-06-18 03:21:50.585	4624	SYSTEM
2026-06-18 03:21:50.585	4672	SYSTEM
2026-06-18 03:21:50.585	4672	nteeshe@lakhans8@gmail.com
2026-06-18 03:21:50.585	4624	DESKTOP-16UPKJNS
2026-06-18 03:21:50.585	4624	nteeshe@lakhans8@gmail.com
2026-06-18 03:21:50.585	4624	DESKTOP-16UPKJNS
2026-06-18 03:21:50.585	4624	nteeshe@lakhans8@gmail.com
2026-06-18 03:21:50.585	4672	nteeshe@lakhans8@gmail.com
2026-06-18 03:21:50.585	4624	DESKTOP-16UPKJNS
2026-06-18 03:21:50.585	4624	nteeshe@lakhans8@gmail.com
2026-06-18 03:21:50.585	4624	DESKTOP-16UPKJNS
2026-06-18 03:21:50.585	4672	SYSTEM
2026-06-18 03:21:50.585	4624	DESKTOP-16UPKJNS

Figure 12: showing the attack timeline

8.2 Indicators of Compromise (IOCs):

- High-volume sequential failed login attempts across multiple accounts within seconds
- All 5 employee accounts targeted simultaneously — characteristic of automated credential stuffing
- SMB protocol (port 445) used as attack vector
- Attempts covered both weak and strong passwords equally consistent with wordlist attack

9. Risk Assessment

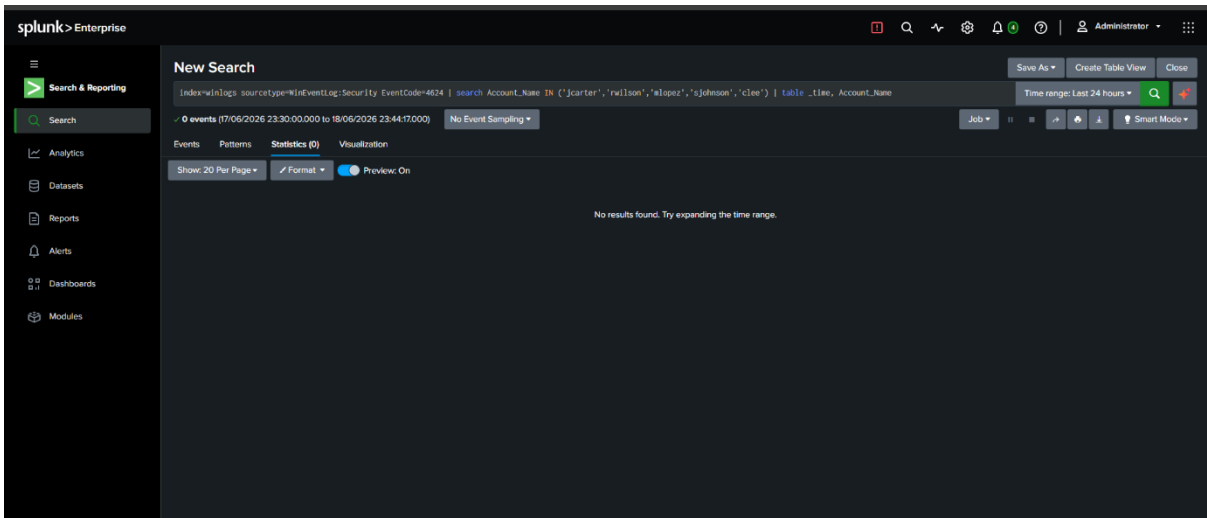


Figure 13: showing 0 results (no successful breach)

9.1 Per-Account Risk Rating:

Account	Department	Failed Logins	Pwd Cracked	Breach	Priv Esc	Risk
Rwilson	HR	24	YES (123456)	No	No	CRITICAL
jcarter	IT	24	YES (Pwd123)	No	No	HIGH
Mlopez	IT	23	No	No	No	MEDIUM
Sjohnson	Finance	23	No	No	No	MEDIUM
Clee	Operations	22	No	No	No	MEDIUM

9.2 Risk Scoring Criteria:

Factor	Low	Medium	High	Critical
Failed Logins	1–4	5–9	10–19	20+
Password Cracked	No	No	Yes	Yes + Trivial
Breach Confirmed	No	No	No	Yes
Privilege Escalation	No	No	Attempted	Succeeded

Overall System Risk Rating: HIGH

Two accounts have crackable passwords, all 5 were actively targeted in a brute-force attack, and SMB authentication is accessible from the internal network. No breach was confirmed during this assessment, but the exposure is significant.

10. Recommendations

10.1 Immediate Actions Required:

- Force password reset for rwilson and jcarter immediately both passwords were cracked in under 20 seconds
- Enforce minimum password length of 14 characters across all accounts
- Enable account lockout policy: lock after 5 failed attempts, 15-minute lockout duration
- Disable SMB access from non-essential internal hosts restrict port 445 via Windows Firewall

10.2 Short-Term Improvements (Within 30 Days):

- Deploy Multi-Factor Authentication (MFA) for all employee accounts
- Implement a password complexity policy banning dictionary words, sequential numbers, and seasonal patterns
- Conduct regular password audits using John the Ripper or similar tools on a monthly basis
- Block known weak passwords using a banned password list (e.g., reject top-10000 common passwords)

10.3 Long-Term Security Improvements:

- Expand Splunk monitoring to include network-level events (DNS, firewall logs)
- Implement a privileged access management (PAM) solution for admin accounts
- Add anomaly-based detection (baseline normal behaviour, alert on deviations)
- Schedule quarterly red team exercises to continuously test detection capabilities
- Deploy endpoint detection and response (EDR) solution for deeper endpoint visibility

11. Conclusion

The SentinelGuard SOC assessment successfully demonstrated an end-to-end security operations workflow: environment setup, endpoint monitoring, identity management, password auditing, threat detection, attack simulation, incident investigation, and risk assessment.

The most significant finding is that 40% of audited accounts (jcarter and rwilson) have passwords that can be cracked within seconds using free, widely available tools. Combined with the confirmed brute-force attack surface via SMB, these accounts represent a high-priority remediation target.

The Splunk-based detection infrastructure successfully identified the attack patterns and generated actionable alerts. With the recommended improvements particularly MFA, account lockout, and password policy enforcement the overall risk posture of this environment would be significantly improved.